AUTONOMOUS SG HACKATHON 2026 | AIRBUS CHALLENGE #3

KASCO

Kalimantan SAF Supply Chain Optimiser

AI-Driven Optimisation for a More Sustainable Aviation Future



Presenter: Wee Zhern Ray

Map shows KASCO's solved base-case network on a Natural Earth coastline.

Event organisers and partners, named for context

Acacia Hackathon Society **NUS Product Club** **Airbus** **Five Degrees** **NUS Hackers** **National University of Singapore**

Scaling SAF is a siting problem as much as a fuel problem

0.6%

SAF share of jet fuel, 2025

80 sites

6,800 kt biomass, challenge data

USD 765M

one large refinery

4 airports

1,200 kt SAF demand

Decarbonisation Aviation needs far more SAF than it has today

Dispersed feedstock Biomass is scattered, and hauling it costs money

Fixed capital A plant locks in site, scale and routes for 20 years

Real interest Airbus and IPB University were reported studying Indonesian biomass logistics and factory siting

Sources: IATA, 9 Dec 2025; QC Intelligence and Biofuels Digest, Aug 2025 (secondary reports); Airbus official update, 31 Oct 2026.

KASCO turns one planning question into an audited network



Mathematical optimisation

Rust solver with a proven lower bound

Geospatial analysis

Interactive map of every site and route

Scenario simulation

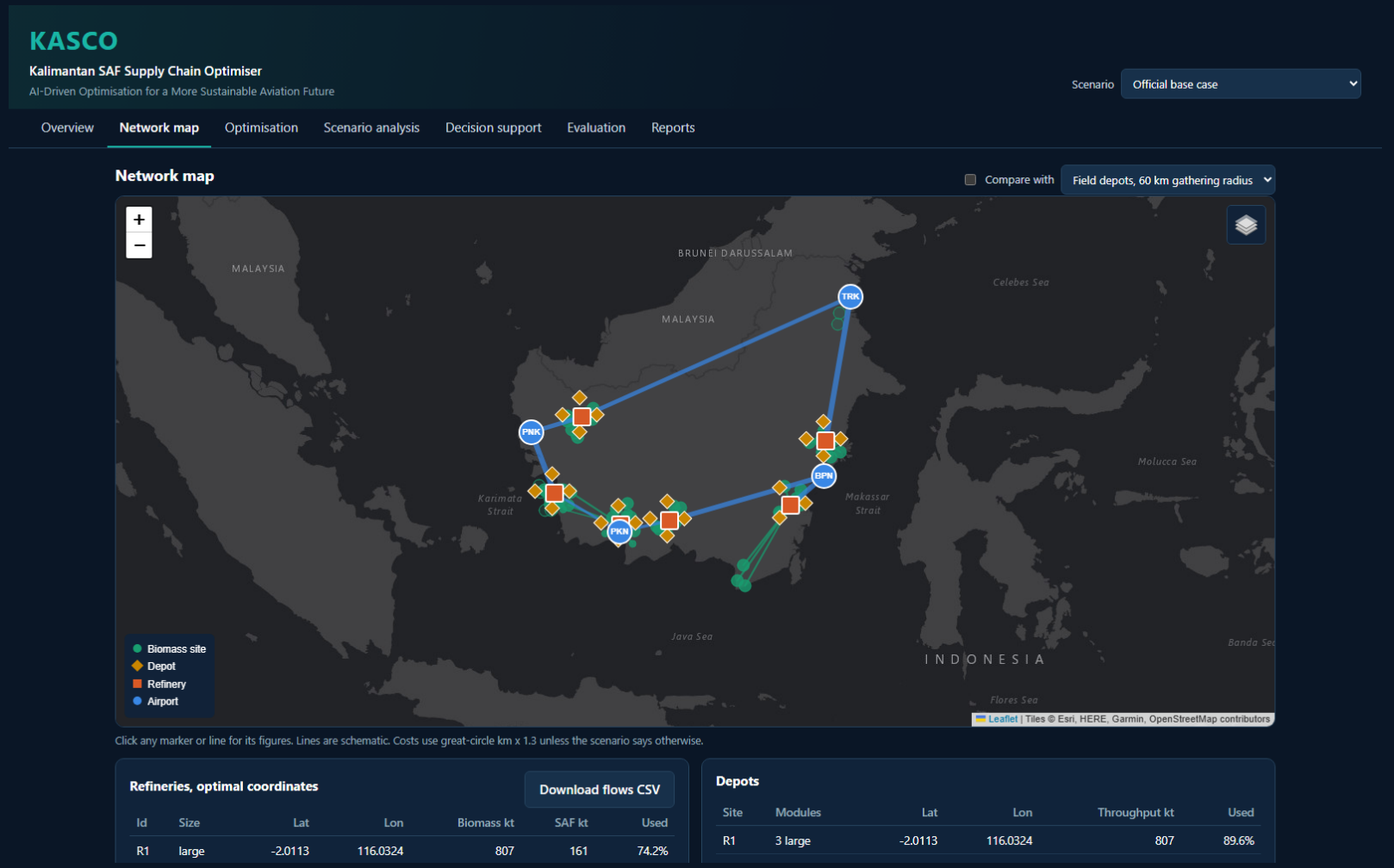
Change inputs and rerun the real solver

Explainable decision support

Answers from solver records, or a refusal

Runs in the browser from GitHub Pages. No server, no API key.

A working prototype, live now



- Interactive map
- Facility selection
- Scenario controls
- Solver outputs
- Explainable answers

weewzr.github.io/airbustrack3ray

Genuine screenshot of the KASCO website, official base case.

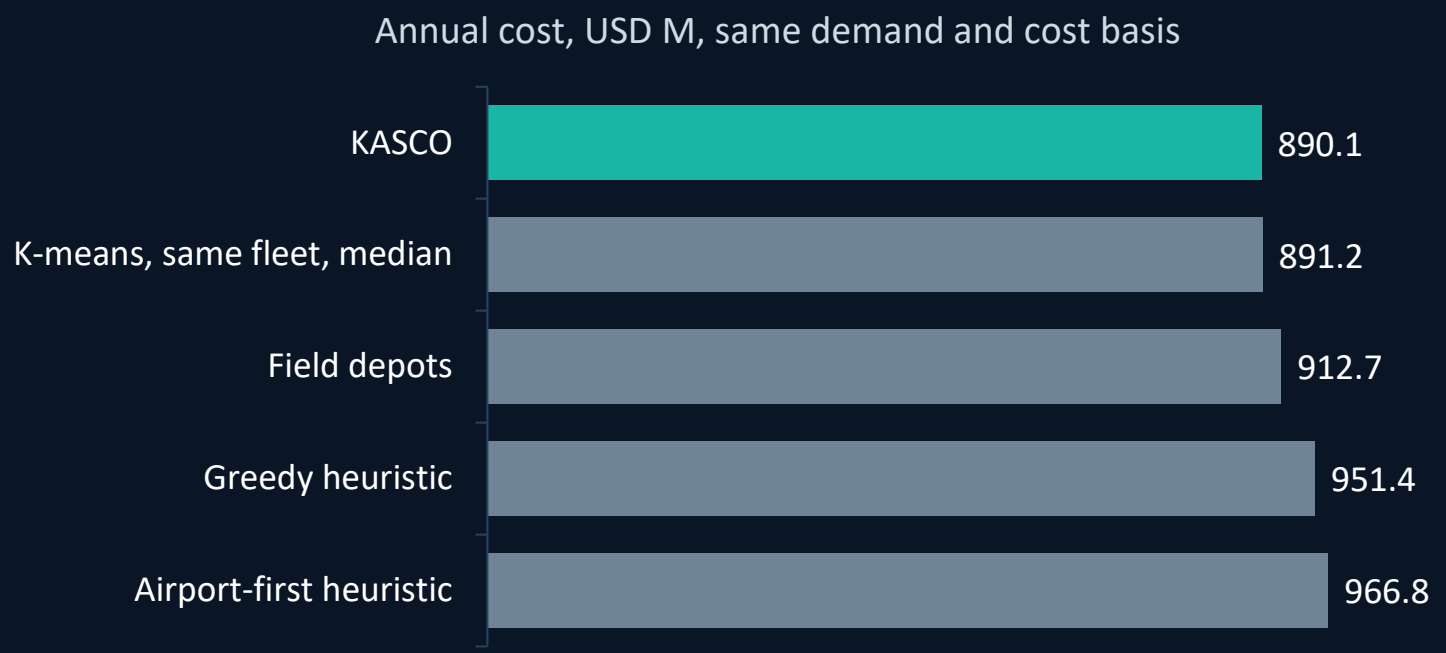
Modelled impact on the official inputs

USD 890.1M
annual cost, modelled

6 + 23
refineries + depot modules

1,200 kt
SAF demand met, 100%

0.75%
proven gap, candidate sites



USD 61M to 77M a year below two simple heuristics, modelled

Same cost from HiGHS, 11 of 11 audit checks pass

K-means given our fleet also reaches this network, so value lies in the proof and limits

Real-world relevance, and its limits

Why it matters

Southeast Asian feedstock could yield about 45.7 Mt of SAF a year by 2050

Fischer-Tropsch ranks among the most viable routes for regional wastes

Indonesia's 1% from 2027 covers Jakarta and Bali departures

Decision limit: past 1,304 kt of SAF, one more kt adds about USD 30M a year

What we do not claim

Official demand is about 12 times today's estimated real jet use at these airports

The 80 biomass points are modelling inputs, and 13 lie off the coast

Sites are screened only: tenure, peat and forest status are unchecked

Depot layout depends on zero mass loss, and real moisture could change it

Sources: RSB and Boeing, Sep 2024; GHD ASEAN study; Fuels and Lubes, Jul 2026; KASCO reality-check scenarios. Full references in the technical report.

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Wee Zhern Ray

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Minor in Hydrogen and Low Carbon Technologies

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From Hackathon MVP to Scalable SAF Decision Intelligence

Stage 1

Current MVP

- Mathematical optimisation
- Interactive geospatial network
- Scenario simulation
- Explainable decision support

Stage 2

AI enhancement

- Secure LLM integration
- Natural-language scenarios
- Possible ML site shortlisting
- Evidence-grounded reports

Stage 3

Real-world validation

- Road and port data
- Industrial site screening
- Feedstock-quality variability
- Robust optimisation, stakeholders

Stage 4

Commercial potential

- Airlines, SAF producers
- Fuel distributors, airports
- Engineering consultancies
- Government planning agencies

Stages 2 to 4 are future possibilities, not established results or commercial achievements.